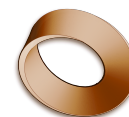


Name: _____

Class: _____

Date: _____



Mr. Rawson • GCSE Physics

Forces 1 Review Packet: Kinematics

Before the Motion Test (10 P Wednesday 30 September, 10 R Thursday 1 October). 20 real exam questions, 65 marks. Every question is on something covered in Forces 1, and every topic here is on the test: scalars and vectors, speed, distance–time graphs, acceleration, velocity–time graphs and $v^2 - u^2 = 2as$.

The questions run from short to long. Each one is labelled with the topics it practises, so use the grid below to go straight to the ones you want. Show every line of working and give every answer a unit; then check against the mark scheme, where each calculation mark is for a line you can point to. Question 16 needs a ruler and a protractor. $v^2 - u^2 = 2as$ is on the AQA equation sheet you get in the test; the other equations you recall.

Topic	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Scalars & vectors	•	•														•				
Speed, $s = vt$			•	•	•	•	•													
Distance–time graphs								•	•	•	•									
Acceleration, $a = \Delta v/t$	•											•	•	•	•					
Velocity–time graphs	•													•	•					
$v^2 - u^2 = 2as$																	•	•	•	•
Marks	3	8	2	2	2	2	2	4	2	4	4	2	4	4	4	3	3	3	3	4

Question 1 | Scalars & vectors 3 marks *Forces PPQ bank Q25(a)(b)*

Some quantities are scalars and some are vectors.

(a) Which of the following quantities are scalars?

Tick (✓) two boxes.

Displacement

☐

Distance

☐

Force

☐

Speed

☐

Velocity

☐

(2)

(b) Give the difference between a vector quantity and a scalar quantity.

(1)

(a) Which of these is a scalar quantity?

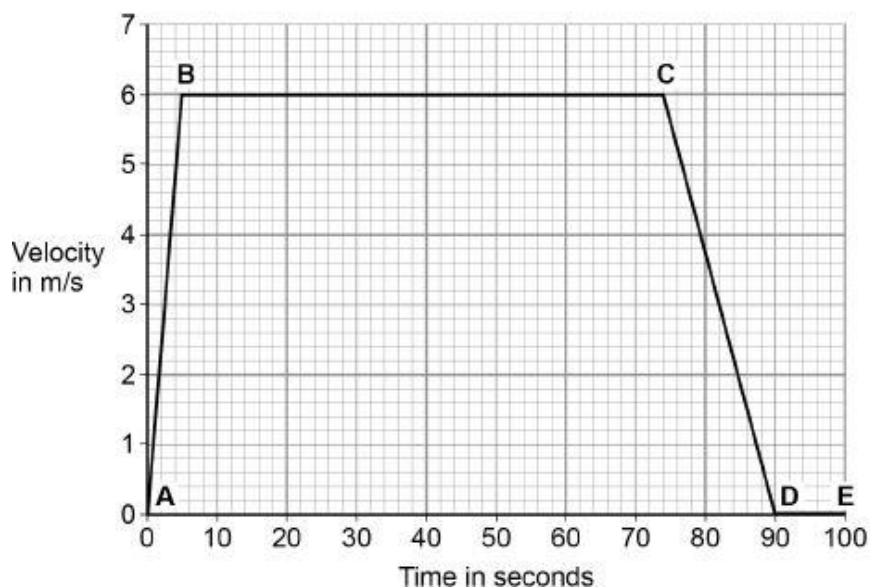
Tick one box.

- | | |
|--------------|--------------------------|
| displacement | ☐ |
| distance | ☐ |
| force | ☐ |
| velocity | ☐ |

(1)

(b) A woman cycled along a straight flat road.

The graph below shows how the woman's velocity changed with time.



Which part of the graph shows the woman moving at constant velocity?

Tick one box.

- | | | | | | |
|----|--------------------------|----|--------------------------|----|--------------------------|
| BC | ☐ | CD | ☐ | DE | ☐ |
|----|--------------------------|----|--------------------------|----|--------------------------|

(1)

(c) Which part of the graph shows the woman stationary?

- | | | | | | |
|----|--------------------------|----|--------------------------|----|--------------------------|
| BC | ☐ | CD | ☐ | DE | ☐ |
|----|--------------------------|----|--------------------------|----|--------------------------|

(1)

Between points A and B the woman was accelerating.

- (d) Use the graph above to determine the total time for which she was accelerating.

Time = _____ s

(1)

- (e) Use the graph above to determine her increase in velocity between points A and B.

Increase in velocity = _____ m/s

(1)

- (f) Calculate her acceleration between points A and B.

Use the equation:

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

Acceleration = _____ m/s²

(2)

- (g) Estimate how a typical cycling speed of 6 m/s compares with a typical walking speed.

Tick one box.

about twice as fast

☐

about four times faster

☐

about eight times faster

☐

(1)

- (a) The wheelchair moves at a constant speed of 2.4 m/s for 4.5 seconds.

Calculate the distance moved by the wheelchair.

Use the equation:

$$\text{distance} = \text{speed} \times \text{time}$$

Distance = _____ m

(2)

- (b) The ball has an average speed of 11 m/s

The ball takes 0.25 s to travel the same distance as the length of the table.

Calculate the length of the table.

Use the equation:

$$\text{distance travelled} = \text{speed} \times \text{time}$$

Length of table = _____ m

(2)

- (c) The ball travels along the lane at an average speed of 4.5 m/s

It takes the ball 4.0 seconds to travel the length of the lane.

Calculate the length of the lane.

Use the equation:

$$\text{distance travelled} = \text{speed} \times \text{time}$$

$$\text{Length of the lane} = \underline{\hspace{2cm}} \text{ m}$$

(2)

- (c) During the first 14 seconds the average speed of the rocket aeroplane on the runway will be 35 m/s.

Calculate the distance that the rocket aeroplane will travel during the first 14 seconds.

Use the equation:

$$\text{distance travelled} = \text{average speed} \times \text{time}$$

$$\text{Distance travelled} = \underline{\hspace{2cm}} \text{ m}$$

(2)

- (b) The car travels a distance of 2040 metres in 2 minutes.

Use the following equation to calculate the mean speed of the car.

$$\text{mean speed} = \frac{\text{distance}}{\text{time}}$$

$$\text{Mean speed} = \underline{\hspace{2cm}} \text{ m / s}$$

(2)

0 2 . 2

The distance–time graph (Figure 2) shows a student walking from school to a friend's house and getting a lift back to school.

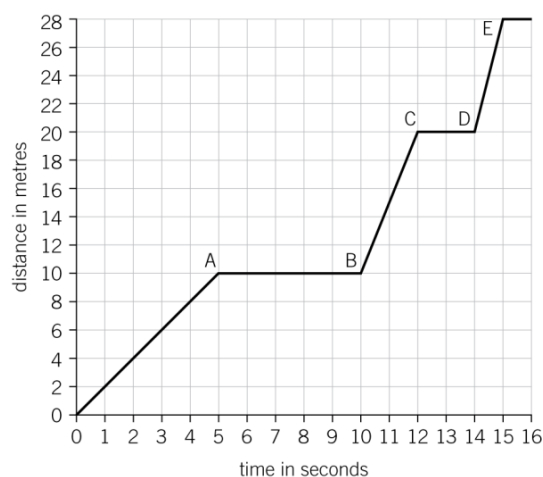


Figure 2

Complete the sentences using words from the box. You may use the words once, more than once, or not at all.

moves uphill	moves at constant speed	moves downhill	stops
--------------	-------------------------	----------------	-------

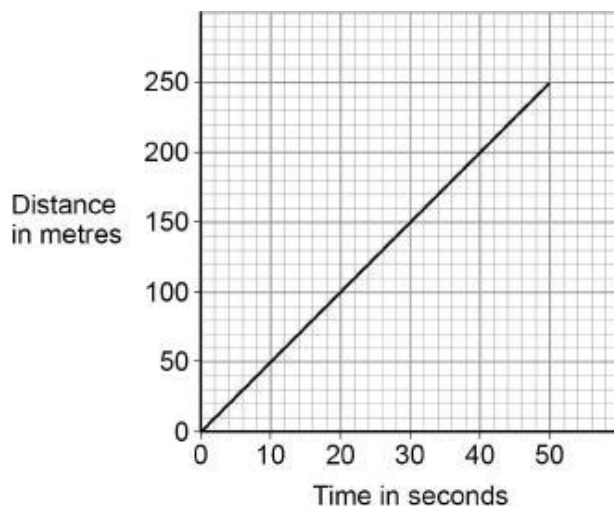
Between A and B, the student

Between B and C, the student

Between C and D, the student

Between D and E, the student (4 marks)

The graph below shows a distance–time graph for 50 seconds of a bicycle ride.



- (a) The gradient of the distance–time graph gives the speed of the bicycle.

Determine the speed of the bicycle.

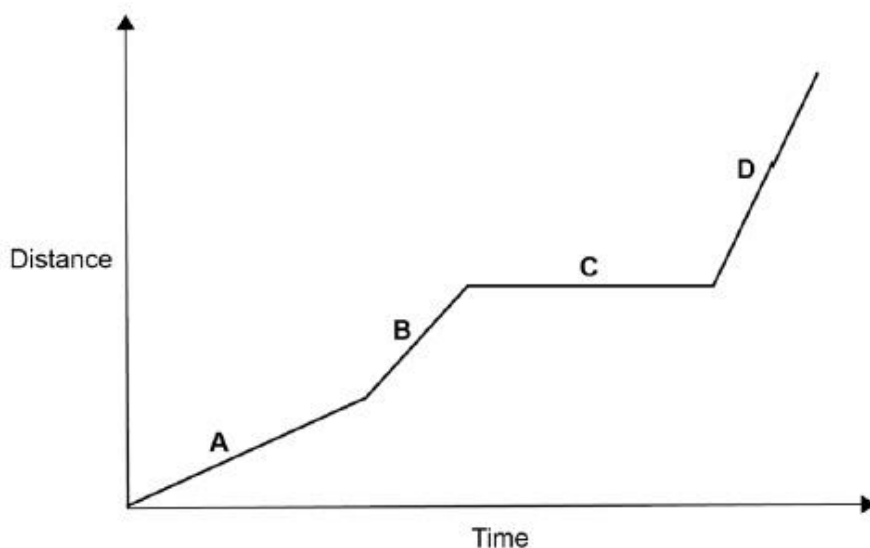
Speed = _____ m/s

(2)

A student measured how the distance travelled by the wheelchair changed over time.

Figure 2 shows a sketch-graph of the results.

Figure 2



- (c) In which section of the graph, A, B, C, or D, did the wheelchair travel fastest?

Give the reason for your answer.

Section _____

Reason _____

(2)

- (d) The student used a data logger with a distance sensor to record the data.

Give two advantages of using a data logger rather than using a stopclock and tape measure.

1. _____

2. _____

(2)

0 1

A small electric truck delivers parcels in a warehouse. The graph of one journey is shown in Figure 1.

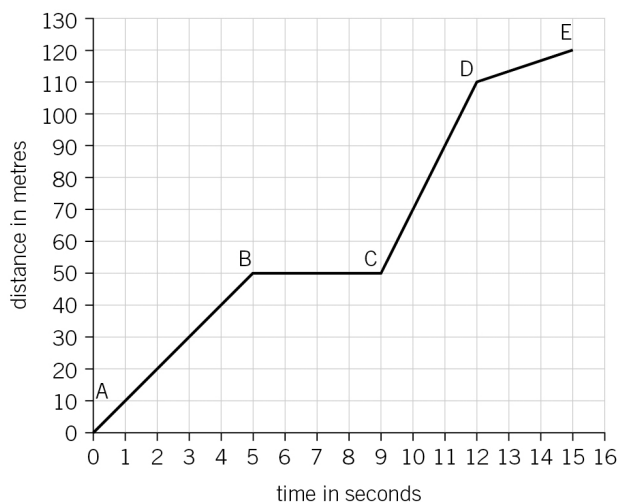


Figure 1

0 1

1

Between which two points in Figure 1 was the truck travelling the fastest? Give a reason for your answer.

.....

(2 marks)

0 1

2

Calculate the speed of the truck between points D and E.

Use the equation:

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

.....

Speed: m/s

(2 marks)

(e) The velocity of the pin changes from 0 to 12 m/s

It takes 0.15 seconds for the velocity to change.

Calculate the acceleration of the pin.

Use the equation:

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

.....

Acceleration = m/s²

(2)

0 3 . 3

In one run of the test, the students recorded the velocity at light gate A as 0.20 m/s, the velocity at light gate B as 0.85 m/s, and the time taken as 2 s.

Calculate the acceleration of the ball from the results obtained during one part of the test.

Give the unit of acceleration. Give your answer to 2 significant figures.

.....

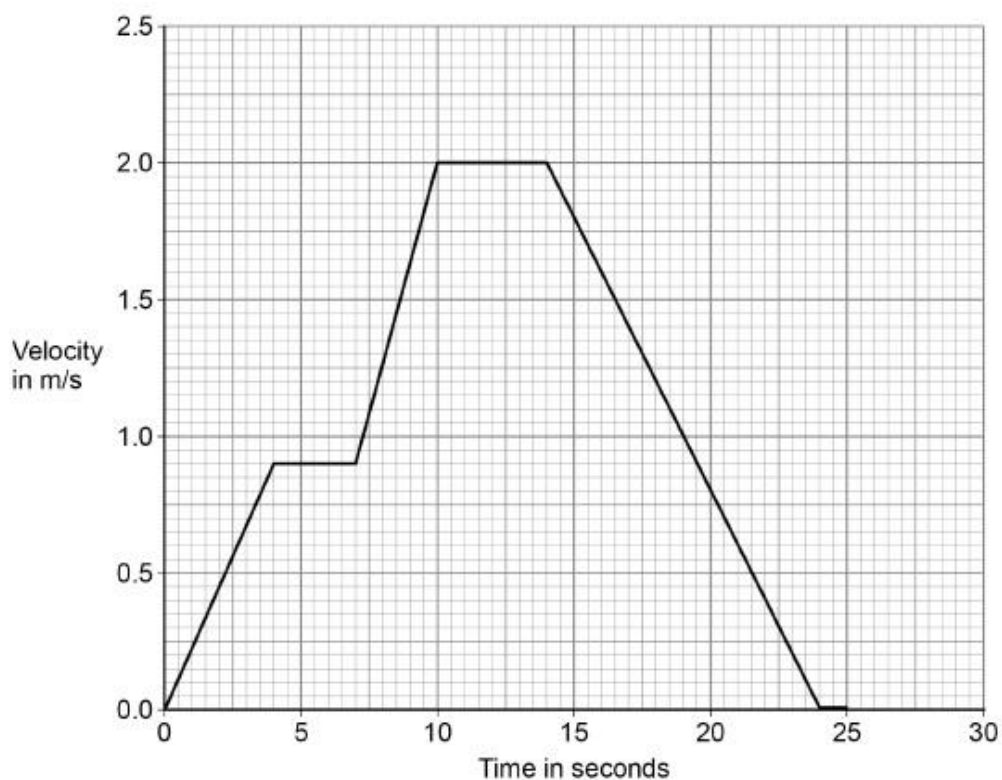
.....

.....

Acceleration : Unit: (4 marks)

Figure 2 is a velocity–time graph for part of the runner’s warm-up.

Figure 2



- (a) Determine the total time for which the velocity of the runner was increasing.

Time = _____ s

(2)

- (b) Determine the deceleration of the runner.

Deceleration = _____ m/s²

(2)

- 0 3 . 4 A second group of students modify the investigation to also record the velocity of the ball as it leaves the end of the tube.

A graph of one test is shown in Figure 4.

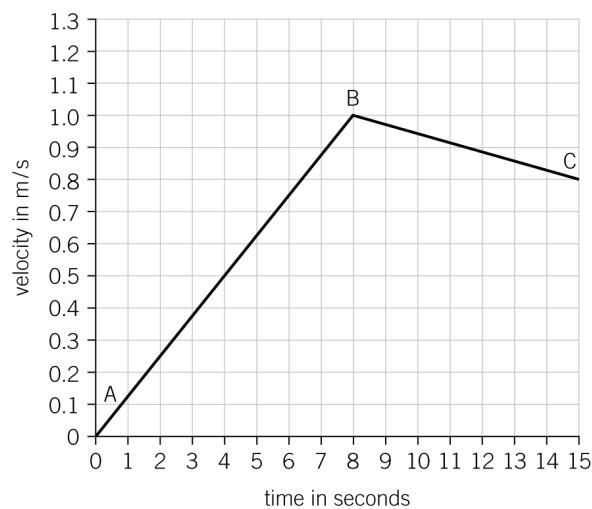


Figure 4

Calculate the acceleration of the ball between points B and C on the graph.

.....

Acceleration :

(3 marks)

- (c) A boat moves through the sea.

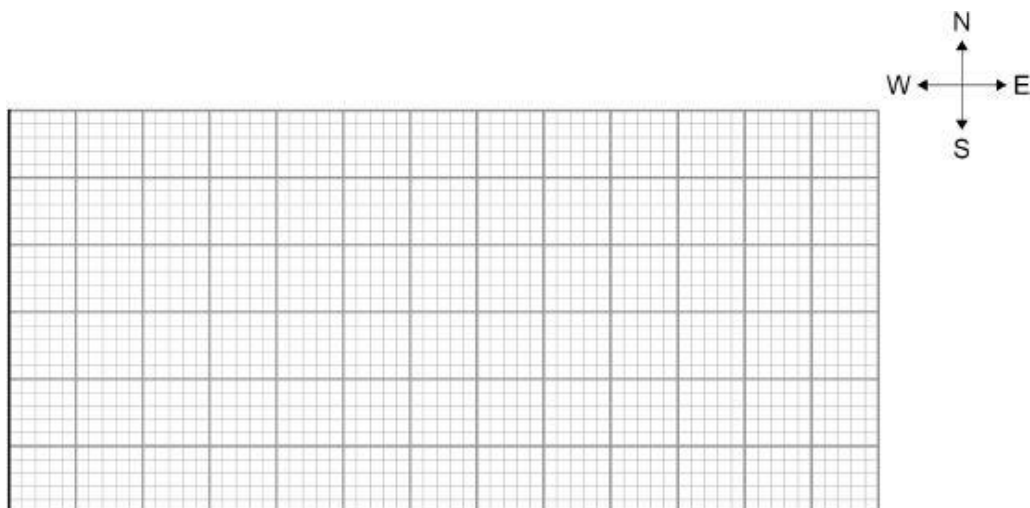
There is a 3000 N force to the west on the boat.

There is a 1000 N force to the south on the boat.

Determine the magnitude and direction of the resultant force on the boat.

Draw a vector diagram of these forces to scale on Figure 2

Figure 2



Magnitude of resultant force = _____ N

Direction of resultant force = _____ °

(3)

- (c) A lorry travels 84 m with a constant acceleration of 2.0 m/s^2 to reach a velocity of 19 m/s

Calculate the initial velocity of the lorry.

Use the Physics Equations Sheet.

Initial velocity = _____ m/s

(3)

- (g) The wheelchair starts from rest.

It accelerates at a constant rate until it has a speed of 1.5 m/s

The wheelchair travels a distance of 2.0 m while it is accelerating.

Calculate the acceleration of the wheelchair.

Using the Physics Equations Sheet.

Acceleration = _____ m/s²

(3)

- (f) A car accelerates at 5.0 m/s² over a distance of 45 m

initial velocity of the car = 0 m/s

Calculate the final velocity of the car.

Use the Physics Equations Sheet.

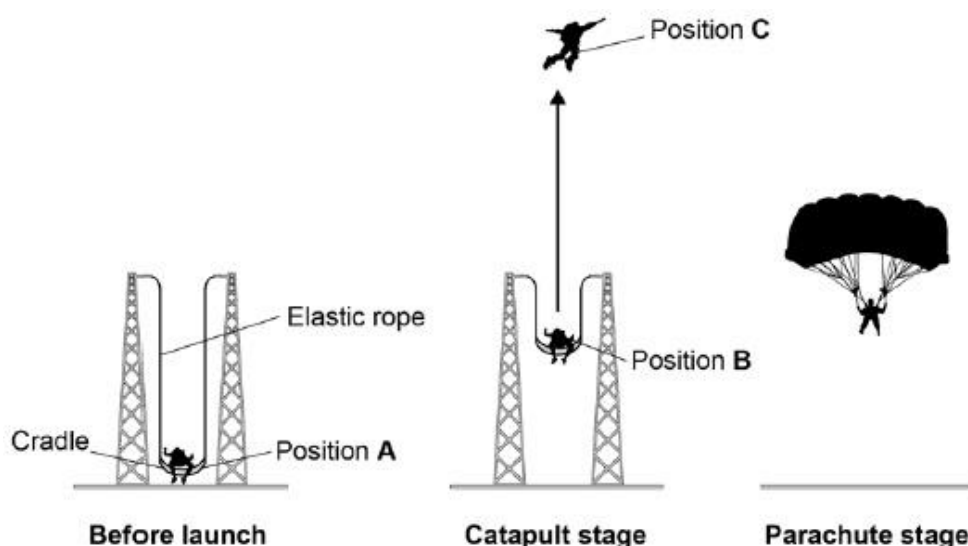
Give your answer to 2 significant figures.

Final velocity (2 significant figures) = _____ m/s

(4)

The diagram shows the stages of an extreme sport called 'human catapult'.

- A person lies in a cradle which is held to the ground.
- The cradle is released.
- The person is launched vertically into the air by an elastic rope.
- The person then parachutes back to the ground.



- (d) The vertical velocity of the person at position B in the diagram is 26 m/s

The vertical velocity at position C is 0 m/s

Calculate the distance between position B and position C. Ignore the effect of air resistance.

Use the Physics Equations Sheet.

Distance = _____ m

(4)

END OF REVIEW PACKET · 65 marks